

Project Charter

PROJECT CHARTER

Project Title	RestoPulse: Restaurant Review Sentiment Analytics System		Project Manager	Juwairiyah Marohombsar	
Project Start Date	April 8, 2026	Project End Date	May 28, 2026	Project Instructor	Jogie Vistal
Business Need					
Restaurant businesses receive a high volume of customer reviews across platforms, making it difficult to manually analyze feedback efficiently. This leads to missed insights, slow decision-making, and inability to detect recurring issues. This project aims to develop a system that automates the analysis of restaurant reviews by applying sentiment analysis and opinion mining techniques. The system will help stakeholders quickly identify customer satisfaction trends, common complaints, and strengths, reducing manual effort and improving data-driven decision-making.					
Project Scope			Deliverables		
Develop a web-based sentiment analytics system that: - Collects and stores restaurant review data (dataset/API/manual input) - Analyzes sentiment (positive, neutral, negative) - Extracts key topics (food, service, ambiance, price) - Displays insights through an interactive dashboard			- Functional sentiment analysis web system - Review dataset (Google Reviews/API or simulated dataset) - Interactive dashboard with charts and insights - Documentation (SRS, system design, diagrams) - Final presentation and system demonstration		
Risks and Issues			Assumptions/Dependencies		
- Limited access to live review data (API restrictions) - Inaccuracy of sentiment classification for complex reviews - Data inconsistency or noise in collected reviews - Time constraints for implementing advanced features			- Review data will be available through dataset or limited API access - Sentiment analysis libraries/tools will function accurately for both English and Filipino text - Users (restaurant owners) will interpret dashboard insights correctly - Team members will complete assigned modules on time		
Overall Concept & Value Proposition					
RestoPulse is a decision-support system that converts unstructured restaurant reviews into clear, actionable insights through sentiment analysis and opinion mining: saving time, revealing key issues and strengths, and enabling data-driven improvements in operations.					

Financials

Budget to complete this project is ₱0 – ₱2000 (minimal, using free tools and datasets)

Milestones Schedule

Milestone	Target Completion Date	Actual Date
Finalize project proposal & dataset	Apr 20, 2026	
Develop backend & sentiment model	May 4, 2026	
Build dashboard interface	May 18, 2026	
System testing & debugging	May 23, 2026	
Final presentation & deployment	May 28, 2026	

Project Team		Approval/Review Committee	
Project Manager & Documentation Lead	Juwairiyah Marohombsar	Project Instructor / Head	Jogie Vistal
UI/UX Designer & QA	Aliyah Salmorin		
Full Stack Dev	Junnel Jonson		
System Analyst	Abdulazis Modi		

WFP

WORK AND FINANCIAL PLAN

Project Title: **RestoPulse: Restaurant Review Sentiment Analytics System**

PROJECT OVERVIEW

RestoPulse is a web-based sentiment analytics system that processes restaurant reviews and converts them into structured insights such as sentiment classification, keyword extraction, and dashboard visualization. The system supports decision-making for restaurant owners by analyzing customer feedback.

Source of Fund: Self-funded (Student Led Contribution)

Implementing Agency: Information Systems Student of College of Information and Computing Sciences

Implementation Site: Class-based development (Laptop/Personal Workstation)

IMPLEMENTATION PERIOD

Start Date: Apr 23, 2026

End Date: May 19, 2026

Total Duration: 26 Days

PLAN & ACTIVITIES

PHASE 1: Project Initiation & Planning

Date: Apr 23 – Apr 27

Target Dates	Activity	Sub-Activity	Item	Cost
Apr 21	Define project scope and objectives	Draft problem statement and do problem analysis (restaurant review inefficiency)		
Apr 22		Define project objectives, scope, and intended outcomes		
Apr 23	Finalize system features and limitations	Feature selection (sentiment, keywords, dashboard)		
Apr 24	Identify tools and technologies	Determine system architecture design		
Apr 25	Prepare project charter document	Draft and organize project charter sections		
		Review and revise charter draft		

PHASE 2: Data Collection & System Design

Date: May 5 – May 8

Target Dates	Activity	Sub-Activity	Item	Cost
May 5	Collect restaurant review dataset (CSV / manual scraping / simulated data)	Gather review entries from dataset/manual sources	Department WiFi	0.00
May 5		Organize reviews into CSV format	Dataset preparation tools	0.00
May 6	Define database structure (SQLite)	Data cleaning plan preparation. Have clean dataset (100–300 reviews)		
May 7		Define sentiment categories (positive, neutral, negative). Do DB schema design		
May 8	Design UI wireframes (dashboard layout)	Identify keywords/topics classification rules	Internet/Data Load (if WiFi isn't available)	100.00
May 8		Draft dashboard wireframe layout		

PHASE 3: System Development

Date: May 9 – May 13

Target Dates	Activity	Sub-Activity	Item	Cost
May 9	Backend development (Flask)	Build backend routes and application structure	Software tools (free/open-source)	0.00
May 10	Sentiment analysis implementation (VADER / NLTK)	Text preprocessing (cleaning, tokenization)	Internet/Data Load (if WiFi isn't available)	100.00
May 11		Sentiment classification module. Accomplish functional sentiment engine		
May 12	Database integration (SQLite)	Keyword extraction logic	System testing utilities	50.00
May 13	Frontend dashboard development (HTML/CSS/JS/Bootstrap)	Chart integration (Chart.js)	Internet/data load	100.00
May 13		Connect dashboard with backend outputs		

PHASE 4: Testing & Improvement

Date: May 14 – May 16

Target Dates	Activity	Sub-Activity	Item	Cost
May 14	Functional testing (input/output validation)	Validate review input and dashboard output flow		
May 14	Sentiment accuracy checking	Test and validate outputs. Compare sentiment output with expected labels	Printing test results	15.00
May 15	UI/UX usability testing	Dashboard clarity enhancements	Department WiFi	0.00
May 15		Error handling improvements		
May 16	Debugging and system refinement	Optimization of sentiment results	Printing test results	15.00
May 16		Final dashboard refinement and fixes (Stable)		

PHASE 5: Finalization & Presentation Preparation

Date: May 17 – May 19

Target Dates	Activity	Sub-Activity	Item	Cost
May 17	Final documentation completion (SRS, diagrams, report)	Compile and finalize project documents	Printing final documents	100.00
May 17		Review and polish documentation		
May 18	Slide preparation for defense	Create presentation deck / materials	Presentation materials	0.00
May 18	System demo preparation	Prepare demonstration scenario		
May 19	Final review and polishing	Rehearsal of defense script		
May 19		Final bug fixes		

SUMMARY OF ESTIMATED PROJECT COST

Phase	Cost
Phase 1	₱0.00
Phase 2	₱100.00
Phase 3	₱250.00
Phase 4	₱30.00
Phase 5	₱100.00
TOTAL	₱480.00

COUNTERPART CONTRIBUTION

Contributor	Contribution
Student Developer	Development time, coding, system design
Institution	Department WiFi access (as needed), academic supervision
Online Resources	Open-source libraries (Flask, NLTK, VADER, Chart.js)
Dataset Sources	Free datasets / simulated review data

This project follows a structured and time-bound implementation plan, making sure that each phase contributes directly to the development of a functional sentiment analytics system within a realistic academic timeframe. The entire system is designed to be **low-cost, lightweight, and feasible**, using open-source tools and limited dataset scope, for completion within the 26-day timeline.

Project Concept

Sentiment Analysis

Input: Text

Process: Analyze words/emotions

Output: Positive / Negative / Neutral

Real-life uses:

Product reviews → Know if customers are happy

Social media → See public opinion about trends

Businesses → Improve services based on feedback

Politics → Gauge public reactions

RestoPulse: Restaurant Review Sentiment Analytics System

Field

- Information Systems + Data Analytics
- Applied AI (NLP for business insights)

Core Problem

Restaurants receive many reviews but:

- Hard to **summarize feedback quickly**
- Owners don't see **patterns (service vs food vs price)**
- Manual reading = inefficient

Solution Focus

A **web-based dashboard system** that:

- Collects restaurant reviews
- Analyzes sentiment (positive/neutral/negative)
- Extracts key themes (e.g., "slow service", "great taste")
- Displays insights visually

👉 Focus: **Decision support for restaurant owners**

🧠 APPROACH

1. Data Source Strategy

- Use **Google Reviews (via scraping or dataset)**
- OR simulated dataset (for safety)

⚠ About Google Reviews API:

- There is no full free public "Google Reviews API"
- If use: Google Places API (limited reviews)

✓ Best approach for a student project:

- Use:
 - Pre-collected dataset (Kaggle-style) OR
 - Small scraped dataset (ethical + limited)

👉 **Perfectly acceptable** as long as you explain limitations.

2. System Flow

Input → Processing → Output

- **Input:** Restaurant reviews (text, rating)
- **Processing:**
 - Sentiment Analysis (NLP)
 - Keyword extraction
 - Category tagging (food, service, ambiance)
- **Output:**
 - Dashboard (charts, insights, summaries)

3. Tech Stack (Student-Friendly)

- **Frontend:** HTML/CSS/JS or React
 - **Backend:** Python (Flask)
 - **NLP:** TextBlob or VADER
 - **Database:** SQLite / Firebase
 - **Visualization:** Chart.js
-

PROJECT CHARTER

1. Project Overview

RestoPulse is a sentiment analysis system designed to help restaurant owners understand customer feedback through automated review analysis and visualization.

2. Objectives

- Automatically classify reviews into sentiment categories
- Identify common issues and strengths
- Provide visual dashboards for decision-making
- Reduce manual review analysis time

3. Scope

✅ In Scope:

- Review collection (dataset/API/manual input)
- Sentiment classification
- Keyword/topic extraction

- Dashboard visualization
- Basic filtering (date, rating, sentiment)

✗ Out of Scope:

- Real-time live API integration (optional only)
- Advanced deep learning models (keep simple)
- Multi-language translation (optional stretch)

4. Key Features

-  Sentiment Score Dashboard
-  Opinion Breakdown (food, service, etc.)
-  Keyword Insights (top complaints/praises)
-  Trends Over Time
-  Rating vs Sentiment comparison

5. Stakeholders

- **Primary Users:** Restaurant owners/managers
- **Developers:** Student team
- **Evaluators:** Professors / panel

The primary stakeholders of RestoPulse are the restaurant owner as the main end user, the project team as system developers, and the faculty adviser as the academic sponsor. Secondary stakeholders include restaurant customers as the data source and project evaluators as system reviewers.

6. Deliverables

- Working web system
- Dataset (or API integration)
- Documentation (SRS, diagrams)
- Final presentation with demo

7. Timeline (Simple)

- Week 1–2: Planning + Dataset
- Week 3–4: Backend + NLP
- Week 5: Dashboard UI
- Week 6: Testing + polish

8. Success Metrics

- ≥80% sentiment classification accuracy
- Clear dashboard visualization
- Functional filtering & insights

- Positive evaluator feedback

9. Risks & Mitigation

Risk	Solution
Limited API access	Use dataset backup
Inaccurate sentiment	Use proven libraries
Data inconsistency	Clean dataset
Time constraints	Limit features to core

SMART FRAMING

“A decision-support system that transforms unstructured customer feedback into actionable business insights.”

EXTRA EDGE (for +10% grade)

- “Top 3 Issues Detected” auto-summary
- Comparison between restaurants
- “Recommended Action” insights

FINAL VERDICT ON GOOGLE REVIEWS

✓ YES, it's fine to use **Google-based data**. BUT:

- Don't rely fully on live API
- Use **dataset + small API sample**
- Clearly explain limitations

Charter Details

PROJECT CHARTER

Project Title: RestoPulse: Restaurant Review Sentiment Analytics System

Business Need

Restaurant businesses receive a high volume of customer reviews across platforms, making it difficult to manually analyze feedback efficiently. This leads to missed insights, slow decision-making, and inability to detect recurring issues.

This project aims to develop a system that automates the analysis of restaurant reviews by applying sentiment analysis and opinion mining techniques. The system will help stakeholders quickly identify customer satisfaction trends, common complaints, and strengths, reducing manual effort and improving data-driven decision-making.

Project Scope

Develop a web-based sentiment analytics system that:

- Collects and stores restaurant review data (dataset/API/manual input)
- Analyzes sentiment (positive, neutral, negative)
- Extracts key topics (food, service, ambiance, price)
- Displays insights through an interactive dashboard

Deliverables

1. Functional sentiment analysis web system
2. Review dataset (Google Reviews/API or simulated dataset)
3. Interactive dashboard with charts and insights
4. Documentation (SRS, system design, diagrams)
5. Final presentation and system demonstration

Risks and Issues

1. Limited access to live review data (API restrictions)
2. Inaccuracy of sentiment classification for complex reviews
3. Data inconsistency or noise in collected reviews
4. Time constraints for implementing advanced features

Assumptions/Dependencies

1. Review data will be available through dataset or limited API access

2. Sentiment analysis libraries/tools will function accurately for English text
3. Users (restaurant owners) will interpret dashboard insights correctly
4. Team members will complete assigned modules on time

Financials

Budget to complete this project is **\$0–\$50** (minimal, using free tools and datasets)

Milestones Schedule

Milestone	Target Completion Date	Actual Date
Finalize project proposal & dataset	1 Week	
Develop backend & sentiment model	2 Weeks	
Build dashboard interface	2 Weeks	
System testing & debugging	1 Week	
Final presentation & deployment	1 Week	

Project Team

Role	Name
Project Manager	[Your Name]
Developer	[Member 1]
UI/UX Designer	[Member 2]
Data Analyst	[Member 3]

Approval/Review Committee

Role	Name
Sponsor	[Instructor Name]

Business Division Head [Panel/Instructor]

Business Unit Head [Panel/Instructor]

Finance Manager N/A

Concept Summary

RestoPulse is a **decision-support system** that transforms unstructured restaurant reviews into structured insights using sentiment analysis and opinion mining.

How It Works (Simple Flow)

1. Input

- Reviews from Google/dataset/manual entry

2. Processing

- Sentiment classification (positive/neutral/negative)
- Keyword extraction (e.g., “slow service”, “delicious food”)
- Categorization (food, service, ambiance, price)

3. Output

- Dashboard with:
 - Sentiment distribution
 - Top complaints & praises
 - Trends over time
 - Rating vs sentiment comparison

Core Value Proposition

- Saves time analyzing reviews
- Provides clear business insights
- Helps restaurants improve operations
- Supports data-driven decisions

Positioning Line

“This system bridges the gap between raw customer feedback and actionable business intelligence by automating sentiment interpretation and insight generation.”

Emphasize:

- **Problem** → **Manual review is inefficient**
- **Solution** → **Automated intelligent dashboard**
- **Impact** → **Better decisions, improved service**

Tech Framework

Feasible Tech Frameworks / System Foundation for RestoPulse

A simple, realistic, and achievable stack.

1. Frontend (User Interface)

Used for dashboard display and user interaction.

Feasible Options

1. **HTML, CSS, JavaScript** (*Recommended*)
 - simplest and easiest
 - lightweight
 - enough for dashboards
2. Bootstrap
 - faster UI design
 - responsive layout
 - good for dashboards
3. React (*Optional, only if confident*)
 - modern UI
 - more scalable
 - but more complex for mini project

Best Recommendation

HTML + CSS + JavaScript + Bootstrap

This is the safest and most practical.

2. Backend (System Logic)

Handles review processing, system flow, and communication with frontend.

Feasible Options

1. Flask (*Recommended*)
 - lightweight
 - beginner-friendly
 - easy API integration
 - ideal for mini systems
2. Django
 - powerful
 - but too heavy for small scope

Best Recommendation

Flask

Best balance of simplicity and capability.

3. Sentiment Analysis / NLP Engine

Core engine for review interpretation.

Feasible Options

1. TextBlob (*Recommended*)
 - simple polarity scoring
 - easy to implement
 - beginner-friendly
2. VADER (*Highly Recommended*)
 - better for review sentiment
 - strong for short opinion text
 - widely used in sentiment projects
3. NLTK
 - useful for preprocessing
 - tokenization, stopwords, cleaning

Best Recommendation

VADER + NLTK

This is your strongest practical combo.

4. Database (Data Storage)

Stores review text, sentiment results, and processed outputs.

Feasible Options

1. **SQLite** (*Recommended*)
 - simple
 - lightweight
 - no server setup
 - perfect for mini project
2. MySQL
 - more scalable
 - but unnecessary for mini scope
3. Firebase
 - useful but overkill for this project

Best Recommendation

SQLite

Most practical and easiest to manage.

5. Data Visualization (Dashboard Charts)

Displays insights in graphs and summaries.

Feasible Options

1. Chart.js (*Recommended*)
 - simple
 - clean visuals
 - easy integration
2. Plotly
 - powerful
 - slightly heavier

Best Recommendation

Chart.js

Simple, clean, presentation-friendly.

6. Data Source / Review Input

Where reviews come from.

Feasible Options

1. **Prepared dataset (CSV/Excel)** (*Recommended*)
 - safest
 - easy to clean
 - reliable for testing
2. **Manual review input**
 - useful fallback
 - easy for demo testing
3. Google Places API (*Supplement only*)
 - limited access
 - not ideal as primary source

Best Recommendation

Prepared CSV dataset + optional manual input

Safest for approval and testing.

7. Development Environment

Where the system is built.

Feasible Options

1. Visual Studio Code (*Recommended*)
2. Jupyter Notebook (*for model testing only*)

Best Recommendation

VS Code + Jupyter (for testing sentiment only)

8. Deployment / Demo

How the project is presented.

Feasible Options

1. **Localhost demo** (*Recommended*)
 - simplest
 - safest for defense
2. Render (*Optional*)
3. PythonAnywhere (*Optional*)

Best Recommendation

Localhost demo

Most reliable during presentation.

Best Overall Recommended Stack

RestoPulse Tech Foundation

- **Frontend:** HTML, CSS, JavaScript, Bootstrap
- **Backend:** Flask
- **NLP Engine:** VADER + NLTK
- **Database:** SQLite
- **Charts:** Chart.js
- **Data Source:** CSV Dataset + Manual Input
- **Development:** VS Code
- **Deployment:** Localhost

The system uses lightweight, beginner-friendly, and proven technologies to have practical implementation, maintainability, and successful completion within the project scope.

Requirements

Project Requirements

1. **Functional Requirements** (*what the system must do*)
2. **Non-Functional Requirements** (*how the system should perform*)
3. **Technical Requirements** (*what is needed to build it*)

1. Functional Requirements

1. The system must accept restaurant review data as input.
2. The system must store review records in a database.
3. The system must preprocess review text (cleaning/tokenization).
4. The system must classify reviews into positive, neutral, or negative sentiment.
5. The system must identify review topics (food, service, ambiance, price).
6. The system must extract frequent keywords from reviews.
7. The system must display sentiment results in a dashboard.
8. The system must visualize trends using charts.
9. The system must allow filtering of reviews by sentiment or category.
10. The system must generate summary insights for decision-making.

“The system’s core function is to collect, process, classify, and visualize restaurant reviews into usable business insights.”

2. Non-Functional Requirements

1. The system must be easy to use and understandable for non-technical users.
2. The system must produce sentiment outputs within acceptable processing time.
3. The system must display clear and readable dashboard visuals.
4. The system must maintain consistent output for repeated inputs.
5. The system must achieve acceptable sentiment classification accuracy (target: 80%+).
6. The system must be reliable during testing and demonstration.
7. The system must be maintainable and easy to update.

“The system is designed to be simple, reliable, readable, and practical for end users.”

3. Technical Requirements

Software Requirements

- Python

- Flask
- HTML/CSS/JavaScript
- Bootstrap
- SQLite
- VADER / NLTK
- Chart.js
- VS Code

Hardware Requirements

- Laptop or desktop
- Minimum 8GB RAM
- Stable internet (for setup only)
- Local storage for dataset and system files

Data Requirements

- Restaurant review dataset (CSV format)
- At least 100–300 review entries for testing
- Review fields:
 - review text
 - rating
 - date (optional)
 - source (optional)

“The project requires only lightweight software, a standard development machine, and a structured review dataset, making it practical and low-cost to implement.”

4. User Requirements

These define who will use the system and what they need.

Primary User

- Restaurant owner / manager

User Needs

1. View customer sentiment summary
2. Identify common complaints and strengths
3. Monitor service trends

4. Use insights for operational decisions

Approval-friendly line

“The primary user is the restaurant owner, who needs fast and understandable insight from customer feedback.”

5. Testing Requirements

These define what must be validated before completion.

1. Functional testing of all core modules
2. Sentiment accuracy validation
3. Dashboard usability testing
4. Input/output consistency checking

Approval-friendly line

“The system will be validated through functional, accuracy, and usability testing to ensure practical and reliable output.”

Overall the project requires three core foundations: a structured review dataset, a lightweight sentiment processing system, and a usable dashboard interface that converts customer feedback into actionable insights.

System Design

System Design RestoPulse: Restaurant Review Sentiment Analytics System

1. User Needs Analysis

Primary User

- Restaurant Owner / Manager

Secondary Users

- Project Administrators / Developers

1.2 User Problems

Restaurant owners often receive numerous customer reviews from online platforms. However, manually analyzing these reviews is time-consuming and inefficient. Because of this:

- Important customer concerns may be overlooked
- Common complaints become difficult to identify
- Service improvement decisions are delayed
- Feedback trends are not easily visible

1.3 User Needs

The users need a system that can:

1. Automatically analyze restaurant reviews
2. Identify whether feedback is positive, neutral, or negative
3. Detect commonly mentioned concerns and praises
4. Present insights in a clear dashboard
5. Help support operational and service decisions

1.4 Proposed Solution

RestoPulse addresses these needs by providing a web-based sentiment analytics dashboard that converts restaurant reviews into structured and actionable business insights.

The system focuses on:

- Sentiment classification
- Opinion/topic identification
- Keyword extraction
- Visual insight reporting

2. Process Mapping

2.1 System Workflow

Step 1 — Data Input

Restaurant review data is collected through:

- CSV Dataset Upload
- Manual Review Input
- Optional API-based review samples

Step 2 — Data Preprocessing

The system cleans and prepares review text by:

- Removing unnecessary symbols
- Converting text to lowercase
- Removing stopwords
- Tokenizing review content

Step 3 — Sentiment Analysis

The system analyzes each review using NLP sentiment analysis tools.

Output Categories:

- Positive
- Neutral
- Negative

Step 4 — Opinion Mining

The system identifies recurring topics and keywords such as:

- Food
- Service
- Ambiance
- Price

Step 5 — Data Visualization

Processed results are displayed in the dashboard through:

- Pie charts
- Bar graphs
- Sentiment summaries
- Trend analysis
- Keyword highlights

Step 6 — Decision Support

Restaurant owners use the generated insights to:

- Improve customer service
- Address recurring complaints
- Monitor customer satisfaction trends
- Support business decisions

2.2 Input-Process-Output Mapping

Input	Process	Output
Restaurant Reviews	Text Cleaning	Cleaned Review Data
Review Dataset	Sentiment Analysis	Sentiment Classification
Customer Feedback	Opinion Mining	Keyword/Topic Insights
Ratings & Reviews	Data Visualization	Dashboard Reports

3. UI Wireframe Components

3.1 Login Page - Allows authorized users to access the dashboard

- System Logo
- Username Field
- Password Field
- Login Button

3.2 Dashboard Page - Displays summarized customer feedback insights

- Navigation Sidebar
- Sentiment Summary Cards
- Pie Chart for Sentiment Distribution
- Top Keywords Section
- Review Trend Graph
- Recent Reviews Panel

3.3 Review Management Page - Allows users to manage and monitor review data

- Upload Dataset Button
- Review Table
- Search and Filter Feature
- Sentiment Labels

3.4 Analytics Page - Provides detailed analytics and insight visualization

- Category Analysis
- Keyword Frequency Graph
- Positive vs Negative Comparison
- Time-Based Trend Reports

Suggested Wireframe Layout

Dashboard Layout Structure

Header

- System Title
- User Profile

Sidebar

- Dashboard
- Reviews
- Analytics
- Reports
- Logout

Main Content

- Sentiment Overview Cards
- Graphs and Charts
- Review Insights
- Recommendation Section

4. Database Structure

The database stores restaurant reviews, sentiment results, and extracted analytics data.

Database Type: SQLite

4.2 Main Tables

Table 1 — Users

Field Name	Data Type	Description
user_id	Integer	Primary Key
username	Text	User login name
password	Text	User password
role	Text	User role

Table 2 — Reviews

Field Name	Data Type	Description
review_id	Integer	Primary Key
review_text	Text	Customer review
rating	Integer	Customer rating
sentiment	Text	Sentiment result
category	Text	Review topic/category
date_added	Date	Date of review

Table 3 — Keywords

Field Name	Data Type	Description
keyword_id	Integer	Primary Key
review_id	Integer	Foreign Key
keyword	Text	Extracted keyword
frequency	Integer	Keyword count

4.3 Relationship Overview

- One review can contain multiple keywords
- Reviews are linked to sentiment classifications
- Users manage uploaded review datasets

4.5. Data Preparation and Sentiment Planning

Data preparation for clean, structured restaurant reviews, and ready for sentiment analysis.

Data Sources

- CSV review datasets
- Manual review input
- Optional API-based review samples (for this kai we may use at least 1 or 3 actual restaurants)

Required Data Fields

Field	Description
review_id	Unique review identifier
review_text	Customer review content
rating	Numerical customer rating
date_added	Review date
source	Source platform

Data Cleaning Process

The system preprocesses reviews before analysis by:

1. Removing special characters and symbols
2. Converting text to lowercase
3. Removing stopwords
4. Removing unnecessary spaces
5. Tokenizing review text

Sentiment Categories

The system classifies reviews into:

- Positive
- Neutral
- Negative

Sentiment Analysis Workflow

Input Review → Text Cleaning → NLP Processing → Sentiment Classification → Dashboard Output

NLP Tools

- VADER - Sentiment scoring
- NLTK - Text preprocessing

Sentiment Scoring Logic

The system analyzes polarity scores to determine review sentiment.

- Positive Score - Positive
- Neutral Score - Neutral
- Negative Score - Negative

Opinion Mining Plan

The system identifies recurring restaurant concerns and praises.

Planned Categories

- Food Quality
- Customer Service
- Ambiance
- Pricing
- Cleanliness

Keyword Extraction

Frequently mentioned words and phrases will be extracted to identify customer trends and recurring feedback.

5. Architecture Design

RestoPulse follows a simple three-layer architecture:

I. Presentation Layer

- HTML
- CSS
- JavaScript
- Bootstrap
- Chart.js

Responsibilities

- Display dashboard interface
- Visualize analytics
- Handle user interaction

II. Application Layer

- Flask Framework
- Sentiment Analysis Engine
- NLP Processing Modules

Responsibilities

- Process review input
- Execute sentiment analysis
- Perform opinion mining
- Generate analytical insights

III. Data Layer

- SQLite Database
- CSV Review Dataset

Responsibilities

- Store review data
- Save sentiment results
- Manage extracted keywords and categories

System Flow Architecture

User → Frontend Dashboard → Flask Backend → NLP Engine → Database → Dashboard Output

6. Technology Stack (Needs Confirmation from Junnel, Developer)

Component	Technology
Frontend	HTML, CSS, JavaScript, Bootstrap
Backend	Flask
NLP Engine	VADER + NLTK
Database	SQLite
Visualization	Chart.js
Development Tool	VS Code

6.5 Backend Logic Flow

Server-Side Process Logic

1. User uploads or inputs review data
2. Flask backend receives review request
3. Backend sends review text to NLP processing module
4. NLP engine cleans and analyzes text
5. Sentiment result is generated
6. Results are stored in SQLite database
7. Dashboard retrieves processed analytics
8. Charts and reports are displayed to users

Frontend and Backend Mapping

Layer	Components	Responsibilities
Frontend	HTML, CSS, JavaScript, Bootstrap, Chart.js	User interface and visualization
Backend	Flask, Python, NLP Modules	Business logic and sentiment processing
Database	SQLite	Data storage and retrieval

API and System Communication Flow

Frontend → Flask Routes → Sentiment Processing Module → SQLite Database → Dashboard Output

7. Testing Plan

7.1 Functional Testing

- Login
- Dataset upload
- Sentiment classification
- Dashboard display
- Filtering functions

7.2 Accuracy Testing to measure correctness of sentiment classification. Target Accuracy:

- 80% acceptable classification accuracy

7.3 Usability Testing

- Interface clarity
- Ease of use
- Dashboard readability

Scope and Limitations

The project focuses on analyzing restaurant reviews from one selected restaurant case study.

The system will:

- Analyze customer reviews
- Classify sentiments
- Generate dashboard insights
- Identify recurring concerns and praises

The system will not:

- Support live real-time review monitoring
- Generate automatic responses
- Perform multilingual sentiment analysis
- Replace full restaurant management systems

Expected Output

The expected output of RestoPulse is a functional web-based analytics dashboard capable of:

- Displaying sentiment summaries
- Highlighting customer concerns and strengths
- Visualizing customer satisfaction trends
- Supporting restaurant decision-making through data-driven insights